A Cosmological Hayden–Preskill Protocol from the FRW Type II_∞ Crossed-Product Algebra

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Abstract

We combine three recent results — (i) the type II_∞ classification of the comoving-diamond algebra of the conformally coupled massless scalar in radiation-dominated FRW [15] (commit `1b2d055`), (ii) the Krylov–Diameter correspondence ($C_k^{-1}\, dC_k/dt = R_{\mathrm{proper}}^{-1}$) on its type II_∞ crossed product [14], and (iii) the Hayden–Preskill scrambling formula [10] — to formulate a *cosmological Hayden–Preskill protocol*: a mathematical analog in which Hubble expansion plays the role of black-hole scrambling. The main statement is: in the past-light-cone (PLC) diamond convention, after a proper time $\Delta t \geq H(t_0)^{-1} \log_2(d_{\mathrm{Hubble}}/k)$, an observer collecting cosmologically expanded radiation can recover an arbitrary $\log_2 k$-qubit message deposited at t_0 with reconstruction fidelity $F \geq 1 - \mathcal{O}(k^2/d_{\mathrm{Hubble}}^2)$, conditional on (a) a rigorous extension of modular Krylov complexity to type II_∞ crossed products and (b) a universal modular Lyapunov exponent $\lambda_L^{\mathrm{mod}} = 2\pi$ on the FRW vacuum. We give a formal protocol description, a proof sketch, explicit comparisons to the original BH scrambling protocol, the Yoshida–Kitaev decoder, and recent strong-pseudorandom-unitary results [6, 16], and an honest assessment of what the result is and is not. The primary intended contribution is conceptual: a cosmological setting in which the Hayden–Preskill bound emerges with no black hole present.

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1 Introduction and statement of the result

The Hayden–Preskill scrambling protocol [10] is one of the cleanest information-theoretic statements about black holes: once a black hole has emitted half of its initial entropy as Hawking radiation, any further qubits thrown in re-emerge in the radiation after the *scrambling time* $t_* = (\beta/2\pi)\log S$. Two ingredients are essential: (i) approximate Haar-random internal dynamics — recently sharpened to *strong* pseudorandom unitaries with logarithmic depth [16, 6] — and (ii) the Bisognano–Wichmann identification $\beta_{\mathrm{mod}} = 2\pi$ of modular flow with thermal flow.

In a different direction, the von Neumann algebra of observables of a gravitational subregion was recently shown to upgrade from type III_1 to type II by tensoring with a clock and crossing with the modular group [18, 4, 5]. For a comoving observer in radiation-dominated FRW the resulting algebra is type II_∞ [15], with a canonical semifinite trace inherited from the Connes–Takesaki dual weight [8]. On this algebra the modular Krylov complexity [2, 17, 1] of the vacuum seed satisfies the geometric identity [14]

$$\frac{1}{C_k(t)}\frac{dC_k(t)}{dt} \;=\; \frac{1}{R_{\mathrm{proper}}(t)} \;+\; \mathcal{O}\big(C_k^{-1}\big), \tag{1}$$

which we call the *Krylov–Diameter correspondence*. In the past-light-cone diamond convention $R_{\mathrm{conf}} = \eta_{\mathrm{obs}}$ with $a(\eta) = a_0\eta$, one has $R_{\mathrm{proper}} = a_0\eta^2 = 2t$, hence

$$\boxed{\frac{1}{C_k(t)}\frac{dC_k(t)}{dt} \;=\; H(t) \;=\; \frac{1}{2t} \quad \text{(radiation FRW, PLC convention)}} \tag{2}$$

exactly at leading order in C_k^{-1}. This is the cosmological analog of the modular Lyapunov saturation $\lambda_L^{\mathrm{mod}} = 2\pi$ of [2], with the Hubble rate playing the role of the modular-thermal frequency.

Our main claim is that (2), combined with the Hayden–Preskill scrambling bound and the cocycle invariance of the modular flow under bounded perturbations [7], yields the following conjectural theorem.

Theorem 1 (Cosmological Hayden–Preskill, conjectural)**.** *Let $\mathcal{A}(D_O^{\mathrm{PLC}})_{\mathrm{FRW}}$ be the comoving past-light-cone diamond algebra of the conformally coupled massless scalar in radiation-dominated FRW ($a(\eta) = a_0\eta$), regularized to a $\log_2 d_{\mathrm{Hubble}}$-qubit finite-trace corner $p\mathcal{N}p$ of the type II_∞ crossed product $\mathcal{N} = \mathcal{A}(D_O^{\mathrm{PLC}})_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R}$, with seed equal to the conformal vacuum Ω_{FRW}.*

Suppose Alice deposits an arbitrary $\log_2 k$-qubit pure state ρ_A into a small subalgebra $\mathcal{M}_A \subset p\mathcal{N}p$ at observer proper time t_0, maximally entangled with a reference R. Suppose Eve applies an arbitrary unitary on the complement $\mathcal{M}_A^c \cap p\mathcal{N}p$. Then, conditional on the two open hypotheses

(H1) *the Block A1 definition of modular Krylov complexity on II_∞ crossed products of [14] §1 is well-posed;*

(H2) *the modular Lyapunov saturation $\lambda_L^{\mathrm{mod}} = 2\pi$ of [2] extends from CFT_2 vacua to the FRW vacuum,*

there exists a Bob-decoder, holding a maximally entangled purification of the remainder of the comoving Hubble volume, that recovers ρ_A from the cosmologically expanded radiation collected over proper time

$$\Delta t \;\geq\; \frac{1}{H(t_0)}\,\log_2\!\Big(\frac{d_{\mathrm{Hubble}}}{k}\Big) \;+\; \mathcal{O}(1/H), \tag{3}$$

with reconstruction fidelity

$$F \;\geq\; 1 \;-\; \mathcal{O}\big(k^2/d_{\mathrm{Hubble}}^2\big). \tag{4}$$

The form of (3) is identical to Hayden–Preskill $t_* = (\beta/2\pi)\log S$ on substituting the cosmological modular temperature $\beta = 2\pi/H(t_0)$. The fidelity bound (4) is the standard Page-curve fidelity [10, 19] adapted to the cosmological corner; both are sympy-verified in `verify.py`.

Structure of the note. §2 sets up the algebra and the comoving-diamond wedge. §3 formalises the protocol. §4 gives the proof sketch, with sympy-verified equations. §5 compares to the BH original, Yoshida–Kitaev, and Brandão–Schuster–Huang. §6 discusses the $\eta_i \to 0^+$ open question and the gauge-conditional caveat. §7 is the section on what this is *not*.

2 Setup: the FRW comoving wedge as a black-hole analog

Fix radiation-dominated flat FRW with line element $ds^2 = a(\eta)^2(-d\eta^2 + d\vec{x}^2)$, $a(\eta) = a_0\eta$. Let O be a comoving observer at $\vec{x} = 0$ with worldline $\gamma_O(t)$. We use two diamond conventions on γ_O:

Past light-cone (PLC) diamond.
D_O^{PLC} is the spacetime region bounded above by the observer's worldline up to current conformal time η_{obs}, and below by the past light cone from $\gamma_O(t)$, truncated at $\eta \to 0^+$. Conformal radius $R_{\mathrm{conf}} = \eta_{\mathrm{obs}}$, proper radius $R_{\mathrm{proper}} = a(\eta_{\mathrm{obs}})\, R_{\mathrm{conf}} = 2t$.

Hubble-horizon diamond.
D_O^{HH} has proper radius $R_{\mathrm{proper}} = 1/H(t)$, used for the de Sitter and matter comparisons of [14].

In what follows, "the diamond" means D_O^{PLC}.

The algebra. Let $\mathcal{A}(D_O^{\mathrm{PLC}})_{\mathrm{FRW}}$ be the von Neumann algebra of the conformally coupled massless scalar field smeared with test functions supported in D_O^{PLC}. By [15, Thm. 3.5], the conformal pullback unitary $U : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{M}}$ implements a W*-isomorphism of $\mathcal{A}(D_O^{\mathrm{PLC}})_{\mathrm{FRW}}$ with the Hislop–Longo [11] algebra of the conformally identified Minkowski diamond, which is type III_1. The crossed product

$$\mathcal{N} \;:=\; \mathcal{A}(D_O^{\mathrm{PLC}})_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R} \tag{5}$$

along the modular automorphism group $\sigma_t^{\omega_{\mathrm{FRW}}}$ is therefore a type II_∞ factor [18], with a canonical semifinite trace $\mathrm{tr}_{\mathcal{N}} = \phi_\omega \circ E$ [8].

The cosmological-BH analog dictionary. Table 1 lists the entries of the analogy.

Black hole (HP07)	**Cosmological analog (this note)**
Schwarzschild horizon	Past-light-cone of comoving observer
Hawking temperature $T_H = 1/\beta$	$T_{\mathrm{cosmo}} = H(t_0)/2\pi$
Scrambling time $t_* = (\beta/2\pi)\log S$	$\Delta t = H^{-1}\log_2(d_{\mathrm{Hubble}}/k)$
Internal Haar-random dynamics	Modular flow $\sigma_t^{\omega_{\mathrm{FRW}}}$
Black hole entropy $S = A/4G$	$\log_2 d_{\mathrm{Hubble}}$ = corner-trace
Outgoing Hawking quanta	Modes redshifted into the past light cone

Table 1: Cosmological–black-hole dictionary.

The substantive non-trivial entry is the *absence* of an event horizon: the comoving wedge has only a cosmological past light cone, not a Killing horizon, so e.g. Faulkner–Speranza [9] do not apply. The crossed-product trace gives us the modular-thermal structure that the Killing horizon would have provided.

3 The protocol

We now state the protocol precisely.

Definition 2 (Cosmological-HP roles)**.** Fix a finite-trace projection $p \in \mathcal{N}$ with $\mathrm{tr}_{\mathcal{N}}(p) = d_{\mathrm{Hubble}}$ (the regularized comoving Hubble dimension at t_0). The corner $p\mathcal{N}p$ is a finite type II_1 subfactor.

- **Alice** is the small subalgebra $\mathcal{M}_A \subset p\mathcal{N}p$ with $\mathrm{tr}_{\mathcal{N}}(p_A) = k$, $p_A \leq p$. Alice deposits a pure state $|\psi\rangle_A \in \mathcal{H}_{\mathcal{M}_A}$ at observer proper time t_0, maximally entangled with a reference R that she keeps.
- **Bob** holds a maximal-entanglement purification of the complement $\mathcal{M}_A^c \cap p\mathcal{N}p$ *at time* t_0. He has access to the cosmologically expanded modes inside the past light cone for $t \in [t_0, t_0 + \Delta t]$.
- **Eve** applies an arbitrary unitary $U_E \in p\mathcal{N}p$ supported on $\mathcal{M}_A^c$ at time t_0. (Equivalently: she perturbs the modular cocycle outside Alice's subalgebra.)

The protocol succeeds if Bob, after receiving the modular-evolved radiation at $t_0 + \Delta t$, can output a state σ_R on R with $F(\sigma_R, |\psi\rangle\langle\psi|_R) \geq 1 - \varepsilon$ for the target ε.

Remark. This is the direct cosmological transcription of HP07's setup: Bob's purification of the complement is the cosmological analog of the "early radiation" Bob has collected from the BH; Eve is the BH internal dynamics; the modular evolution from t_0 to $t_0 + \Delta t$ is the analog of the BH scrambling. The crucial difference is that the modular flow here is geometric (Bisognano–Wichmann after conformal pullback [11, 15]), not stipulated to be Haar.

4 Proof sketch

The proof of Theorem 1 (modulo (H1)/(H2)) decomposes into three steps.

4.1 Step 1: Krylov–Diameter correspondence on the corner

By [14] §§2–3, the modular Krylov spread complexity $C_k^{(p)}(t)$ on the corner $p\mathcal{N}p$ with seed $p\Omega_{\mathrm{FRW}}p$ obeys

$$\frac{1}{C_k^{(p)}(t)} \frac{dC_k^{(p)}(t)}{dt} \;=\; \lambda_L^{\mathrm{mod}} \cdot \frac{ds}{dt}\bigg|_{\mathrm{center}} \;=\; \frac{2\pi}{2\pi\, R_{\mathrm{proper}}(t)} \;=\; \frac{1}{R_{\mathrm{proper}}(t)}, \tag{6}$$

where the second equality uses the universal modular Lyapunov exponent [2] and the Casini–Huerta–Myers [3] local Unruh temperature $T_{\mathrm{loc}} = 1/(2\pi R_{\mathrm{proper}})$ at the diamond center. In the PLC convention $R_{\mathrm{proper}} = 2t$, so (6) reduces to $H(t)$ exactly — this is verified algebraically in `verify.py` (V1, V4).

The integrated form is

$$C_k^{(p)}(t_0 + \Delta t) \;=\; C_k^{(p)}(t_0)\, e^{H(t_0)\,\Delta t} \left[1 + \mathcal{O}\big((H\Delta t)^2 H/H_{\mathrm{Hubble}}\big)\right], \tag{7}$$

where the bracket comes from the slow drift of $H(t) \propto 1/t$ over the scrambling timescale.

4.2 Step 2: Connes 1973 cocycle invariance

By Connes' 1973 cocycle theorem [7], modular flows of faithful semifinite normal weights on a von Neumann algebra are unique up to inner cocycles. In particular, Eve's perturbation U_E on the complement subalgebra changes the modular flow only by an inner cocycle $u_t^{U_E} \in p\mathcal{N}p$ which is contained in Eve's region. This is the operator-algebraic version of the statement that "Eve's local dynamics cannot reach Alice's subalgebra" — a finite-corner version of the Tomita–Takesaki uniqueness used in the QKD-style D1 conjecture of [14, § D1]. On the corner $p\mathcal{N}p$, this gives a quantitative bound: for any $V = V^* \in \mathcal{M}_A^c$ with $\|V\| \leq \varepsilon$,

$$\left\|\sigma_t^{\omega_{\mathrm{FRW}}^{U_E}}|_{\mathcal{M}_A} - \sigma_t^{\omega_{\mathrm{FRW}}}|_{\mathcal{M}_A}\right\| \;\leq\; 2\varepsilon^2 \,\langle K_{\mathrm{FRW}}^2\rangle_{\Omega_{\mathrm{FRW}},p} \;+\; \mathcal{O}(\varepsilon^4), \tag{8}$$

the precise form (D1-EQN) sympy-verified in [14] sub-agent crypto-synthesis script.

4.3 Step 3: Page-curve / HP recovery

With Steps 1 and 2 in place, the standard Hayden–Preskill argument [10, §4] runs verbatim on the type II_1 corner $p\mathcal{N}p$ once we identify:

- "Black hole entropy at deposition time" $\to \log_2 d_{\mathrm{Hubble}}$;
- "Modular Haar scrambling at rate $1/\beta$" $\to$ modular flow at rate $\lambda_L^{\mathrm{mod}} \cdot ds/dt = H(t)$;
- "Half-evaporation point" $\to$ saturation of corner trace.

HP07 eq. (4.6) then yields the bound (4) on the Bob-decoder fidelity, and the saturation condition $C_k^{(p)}(\Delta t) \geq d_{\mathrm{Hubble}}/k$ combined with (7) gives the threshold time (3). `verify.py` (V2, V3) confirms the algebra.

One open math.OA step. The proof is conditional on (H1) (rigorous Block A1 definition of C_k on II_∞ crossed products) and (H2) (modular Lyapunov saturation on the FRW vacuum, not just the CFT_2 vacuum used by [2]). The first is a Connes–Takesaki dual-weight bookkeeping problem; the second is a hep-th statement that should be tractable by direct Lanczos on the Hislop–Longo modular generator on the FRW vacuum, using the spectral density $\rho(\omega) = 1/(2\cosh^2(\pi\omega/2))$ for the conformally coupled weight-1/2 primary — this density, together with $b_n/n \to 1/2$ in seed-normalised units (rescaling to $b_n \to \pi n$ in BW units), is what [14] §2 numerically obtains. The first step is the harder one. □

5 Comparison to existing scrambling protocols

Original Hayden–Preskill (BH). [10] considers a black hole that has emitted half its entropy as Hawking radiation, with internal dynamics modelled as Haar-random unitaries on the joint BH+radiation Hilbert space. The scrambling time $t_* = (\beta/2\pi)\log S = (2\pi/\beta_{\mathrm{Hawking}})^{-1}\log S$ is saturated. Our cosmological version replaces "Haar-random internal dynamics" by "modular flow with universal Lyapunov $\lambda_L = 2\pi$" — i.e. we trade an unmodelled assumption for a physically derived rate (conditional on (H2)). The exponent is the same.

Yoshida–Kitaev decoder. [19] constructs an explicit deterministic decoder for the HP protocol with circuit complexity $\mathcal{O}(d_A C)$ (where C is the circuit size of the scrambler) and a probabilistic decoder with success $1/d_A^2$. Their construction uses Grover-amplified teleportation. In our cosmological setting, the modular flow is generated by the FRW Hamiltonian, so an explicit "Bob-decoder circuit" translates to: prepare Bob's purification of $p\mathcal{N}p \setminus \mathcal{M}_A$, evolve under $-iK_{\mathcal{N}}$ for modular parameter $s = 2\pi H(t_0)\Delta t$, then apply a Petz recovery map for the small-corner inclusion $\mathcal{M}_A \hookrightarrow p\mathcal{N}p$. The Yoshida–Kitaev $\mathcal{O}(d_A C)$ scaling becomes $\mathcal{O}(k \cdot e^{H\Delta t})$, and the $1/d_A^2$ probabilistic-decoder fidelity becomes $1 - \mathcal{O}(k^2/d_{\text{Hubble}}^2)$, matching (4).

Brandão et al. strong pseudorandomness. The very recent results of [6] and [16] sharpen the HP assumption: they show that strong pseudorandom unitaries (robust to inverse and transpose queries) can be generated in optimal logarithmic depth, providing operational support for the fast-scrambling conjecture. Our result is logically orthogonal: we do *not* need a pseudorandomness assumption on the FRW dynamics — the modular flow is fixed by Bisognano–Wichmann via the conformal pullback. Instead, the non-trivial input is the modular Lyapunov saturation $\lambda_L^{\text{mod}} = 2\pi$, which is structurally analogous to the Maldacena–Shenker–Stanford bound saturation [12] and is the modular analog of fast scrambling.

Modular complexity literature. The Krylov-complexity machinery draws directly on Caputa–Magán–Patramanis–Tonni [2] (the universal $\lambda_L^{\text{mod}} = 2\pi$), Vardian [17] (the modular-Krylov definition extending Parker et al. [13]), and Aguilar-Gutierrez [1] (the closest published precedent — DSSYK on II_1). The D_O^{PLC} specialisation is, to our knowledge, new.

6 Limits, caveats, open questions

The Big-Bang touching limit ($\eta_i \to 0^+$). The PLC diamond extends back to the Big Bang. [15] flags three obstructions in 4D: (a) failure of the conformal-rescaling bijection $f \mapsto a^3 f$ at $\eta_i = 0$, (b) IR singularity of the conformal-vacuum two-point function on the diamond past tip, (c) non-Hadamard boundary behaviour. In 2D all three vanish; in 4D none of them are resolved here. The resolution of [15] Open Question 4.4 is a precondition for promoting Theorem 1 from conjecture to theorem in 4D.

Gauge-conditional caveat from Piste 1 audit. [14] (gauge audit, sympy-verified) shows that the substantive content of the Krylov rate identity is $(C_k^{-1}\, dC_k/dt) = R_{\text{proper}}^{-1}$, which is a spacetime scalar on the worldline and thus invariant under any slicing/parametrisation. The match $(C_k^{-1}\, dC_k/dt) = H(t)$ is, however, era- and diamond-convention-conditional:

- Radiation+PLC: ratio 1 (used here).
- Matter+PLC: ratio 1/2.
- de Sitter+Hubble-horizon: ratio 1 (tautological).

Theorem 1 therefore holds in the *(radiation+PLC) sector only*; an analogous statement for matter or dark-energy-dominated FRW would have a different $\mathcal{O}(1)$ prefactor in (3).

Finite-N validity. The whole construction lives on a regularised corner $p\mathcal{N}p$ with $\text{tr}_{\mathcal{N}}(p) = d_{\text{Hubble}} < \infty$. Finite-$N$ numerical checks on the type II_1 DSSYK setup [1] would test (i) the exponential growth rate (7) and (ii) the saturation onset.

Cosmological observability. The protocol is a mathematical analog. There is no proposal here to build a Bob-decoder in the actual universe. The tightest physical comparison would be to the recombination era, where one might in principle ask whether CMB modes can "decode" a small subset of pre-recombination modes at the HP fidelity bound; this is out of scope.

7 What this is NOT

It is intellectually honest to be explicit:

(1) This is not a proven cryptographic primitive. The bound (4) requires Eve's restriction to the complement subalgebra of $\mathcal{M}_A$ at time t_0. We do not exhibit an adversary model in which this restriction is enforced by a physical resource constraint (energy, bandwidth, locality). In the QKD literature this would be called a *restricted-Eve* bound, not a full cryptographic security proof.

(2) This is not a bound on cosmological observables. Theorem 1 says nothing about CMB temperature anisotropies, recombination, or any directly measured cosmological quantity. The "d_{Hubble}" here is a regularisation parameter ($\text{tr}_{\mathcal{N}}(p)$), not a calibrated number of physical degrees of freedom in a Hubble volume. Connecting the two would require a separate argument about the von Neumann dimension of the Hubble corner.

(3) This is not gauge-unconditional. As noted in §6, the result lives in the (radiation+PLC) diamond convention. The analogous statement in matter or de Sitter has a different prefactor; the substantive identity is the geometric R_{proper}^{-1} rate, not the cosmological $H(t)$.

(4) This is not a theorem on II_∞. We rely on two open hypotheses (H1)/(H2). The math.OA hypothesis (H1) is believed to be tractable but has no published treatment for II_∞ crossed products; the closest precedents are CFT_2 [2], II_1 DSSYK [1], and the bulk-reconstruction modular-Krylov of [17]. The hep-th hypothesis (H2) (modular-Lyapunov saturation on the FRW vacuum, not just the CFT_2 vacuum) requires direct Lanczos analysis on the Hislop–Longo modular generator on the FRW vacuum — the numerics in [14] §2 are consistent with it but do not prove it.

(5) This is not a substitute for the original HP07 result. HP07's BH protocol applies to actual evaporating black holes and is a statement about Hawking radiation. The cosmological protocol is an analog: same equation, different physics, no event horizon.

(6) This is not free of the Big-Bang singularity. [15] §4 Open Question 4.4 must be addressed before the $\eta_i \to 0^+$ limit can be promoted from working assumption to theorem in 4D. In 2D the obstructions vanish.

8 Conclusion and target

The combination of (i) the type II_∞ classification of the FRW comoving-diamond crossed-product algebra and (ii) the universal modular Lyapunov $\lambda_L^{\text{mod}} = 2\pi$, plus (iii) the standard Hayden–Preskill recovery argument, gives a clean cosmological analog of black-hole scrambling: (3) is the cosmological-HP recovery time, with Hubble rate playing the role of inverse modular temperature; (4) is the standard Page-curve fidelity bound. Three explicit equations ((2), (3), (4)) and a fourth supporting one ((7)) are sympy-verified.

The result is *conjectural* pending (H1)/(H2), and lives in the (radiation+PLC) sector. The substantive new structural input is the geometric Krylov–Diameter correspondence (1), which is robust under the gauge audit of [14].

Publication target. We see two natural homes:

quant-ph short note (12–16 pp).
Frame the result as a cosmological analog of HP07. Best fit: a quant-ph e-print, possibly Quantum or JHEP. The conditional nature of the theorem must be flagged in the abstract.

Math.OA long paper (40+ pp).
The companion paper that closes hypothesis (H1) — rigorous modular Krylov complexity on II_∞ crossed products via Connes–Takesaki dual weights — is the math.OA-paper material, plausibly Comm. Math. Phys. or J. Funct. Anal.

Our recommendation: write the present note up as a 12-page quant-ph short note, with the math.OA paper as a follow-up. We do *not* recommend submission of the present note as a long paper at this stage: the conditional-on-(H1)/(H2) status is too prominent for the result to stand alone in a flagship venue. PRD is a possible alternative if the hep-th angle is emphasised over the operator-algebra one.

Acknowledgements. The Krylov–Diameter identity emerged from sub-agent computations on `paper/frw_typeII_note/` commit `72c3617`; the gauge audit of that identity (responding to a Mistral large-latest counter-review) sharpened the era/convention caveats. All citation metadata was independently verified via the arXiv API on 2 May 2026. Sympy verification (5 algebraic claims) is in `verify.py`; the script runs end-to-end and passes.

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